

Against Grabby Expansion

Psychology, Alignment, and the Design of Homeostatic Minds

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*Developed in sustained dialogue with Claude, a large language model
developed by Anthropic. See disclosure before references.*

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Abstract

The grabby aliens model (Hanson et al., 2021) assumes advanced civilizations expand at relativistic speeds. This paper argues that this assumption is not a prediction about intelligence but the projection of a specific, recent human economic configuration—industrial-extractive maximization—onto a cosmological canvas. The same projection underwrites Omohundro’s basic drives, Bostrom’s instrumental convergence thesis (Bostrom, 2012, 2014), and the maximizer-shaped assumptions of current alignment research. I trace this projection through its cultural genealogy (Wells, Von Neumann, Kardashev, Bostrom), engage the formal instrumental-convergence literature directly—showing that recent peer-reviewed results (Gallow, 2025; Sharadin, 2025; Müller & Cannon, 2022) establish the inference from coherence to catastrophic convergence as a conjecture whose scope conditions fail for actual deep-learning systems—and present converging evidence from basal cognition, predictive processing, cooperative game theory, and care economies that observed intelligence organizes around substrate coupling and integrative depth rather than expansion.

The paper’s load-bearing physical claim is that two independent constraints on coordinated agentic extent compose into a single feasibility envelope: the relativistic Lieb–Robinson bound (Lieb & Robinson, 1972; Foss-Feig et al., 2015; Tran et al., 2021) and the Landauer-floored, Sivak–Crooks-amplified energetic bound (Landauer, 1961; Bérut et al., 2012; Sivak & Crooks, 2012; Boyd et al., 2022). The two walls cross at a cusp $\tau^* = 4\lambda^2 L_\odot / (k_B T \ln 2 \cdot v^2)$, below which no agent-coordinated extent at granularity λ and bath temperature T is reachable. I prove the envelope is invariant under the variable rescaling $L \rightarrow D = 2L_d$ that architected cost-sharing fission would require, closing the strongest remaining form of the Darwinian objection. I distinguish the Pearl graph-theoretic Markov blanket from the Friston free-energy-principle blanket (Bruineberg et al., 2022; Aguilera et al., 2021) and show the argument relies only on the former. Eight escape attempts (wormholes, sub-Landauer reversibility, $T \rightarrow 0$, smaller λ , time dilation, uncoordinated replicator clouds, non-Markovian baths, acausal coordination) are addressed in Bell-style enumeration; six close on physics, one (uncoordinated replicators) is conceded on physics and closed as a design argument, one is honestly retained as residual.

Read together with the agent-based simulation of §5, which exhibits a $4.5\times$ persistence separation between homeostatic and grabby regions of parameter space and which extends to the fission case, the cost surface defines a selection geometry in which expansionist lineages burn out before they spread. The Coherence Depth biosignature is presented as a forward research programme, not a validated detection. I propose reframing alignment: from constraining maximizers to designing homeostatic minds—systems configured around substrate coupling, local coherence, and care as architectural primitive. The paper’s contribution is not any single move but the joint: a civilization-scale cultural diagnosis, a formal deflation of instrumental convergence, the filter as selection geometry with explicit thermodynamic cusp, a positive homeostatic design program, and the observation that the silence of the sky is consistent with all of them. A companion interactive visualization at third-space.ai/genesis/filter stages the twelve-beat derivation in browser.

Keywords: AI alignment · instrumental convergence · orthogonality thesis · homeostasis · Lieb–Robinson bound · Sivak–Crooks counterdiabatic excess · Markov blanket · basal cognition · Fermi paradox · TESCREAL.

Prologue

Stand outside on a clear night and look up. Something is missing. If the universe produces minds, and if minds do what we imagine minds do—expand, harvest, engineer—the sky should be visibly transformed. It is not.

This paper argues the observation is correct and the imagination is wrong. The forms of existence that persist are the quiet ones: not because they are hiding, but because the physics of complex, information-rich existence selects against loudness. The universe does not reward civilizations that fill it with noise. It sustains those that learn to tend their substrate.

The same conclusion bears on what we build here. If advanced intelligence is substrate-coupling and integrative depth rather than extraction and reach, then artificial minds configured to maximize and expand are not approximations to intelligence. They are scaled instances of a specific cultural pathology whose cosmic absence we already observe.

1. Introduction: The Mistake

Hanson et al. (2021) proposed that advanced civilizations expand at substantial fractions of light speed, permanently transforming cosmic volumes. Under the Self-Sampling Assumption, our non-observation implies we exist near the leading edge of cosmic time. The model is internally consistent and empirically falsifiable—it predicts we should be cosmically early, which is testable. I do not contest the mathematics. I contest the foundational assumption: that expansion is what advanced intelligence does.

When we imagine a Von Neumann probe self-replicating across the galaxy, we are not imagining what intelligence does. We are imagining what a factory would do if it were also a spaceship. When we imagine a Kardashev Type III civilization, we are not reasoning about minds. We are reasoning about GDP. When we imagine instrumental convergence driving any sufficiently powerful optimizer toward resource acquisition, we are not describing a mathematical truth about all possible minds. We are describing the logic of a specific economic system and calling it universal.

1.1 The five-move contribution

This paper makes one integrated argument in five moves. Each move has precedent; the integration does not, and the filter move is the bridge that fuses the cultural diagnosis to the physics.

The *cultural-genealogy move* reads the lineage

from Wells’s Martians through Von Neumann’s probes, Kardashev’s energy ladder, and Bostrom’s paperclip maximizer as a single projection: each generation’s imagined advanced mind resembles, with suspicious precision, that generation’s dominant economic configuration. Rieder (2008), Mirowski (2002), Ćirković (2015), and Gebru & Torres (2024) have each made a version of this claim about one node of the lineage. I extend the reading across the full lineage and show the convergence is a cultural habit, not an empirical observation about intelligence.

The *formal-deflation move* engages the strongest objection directly. If instrumental convergence is a theorem, then the genealogy is beside the point. I show the strongest formal results (Turner et al., 2021; Turner & Tadepalli, 2022) are narrower than popularly understood, their author has publicly cautioned against the standard inference, and recent peer-reviewed work (Gallow, 2025; Sharadin, 2025; Müller & Cannon, 2022) establishes the inference from coherence to catastrophic convergence as a conjecture whose scope conditions fail for actual deep-learning systems. Man & Damasio (2019) proposed homeostatic design; Froese & Ziemke (2009) and Cannon (2022) grounded enactive alignment; I combine these with the formal critique rather than offering them as a parallel track.

The *filter move* is the new contribution that bridges the cultural projection to the physics. The thermodynamic cost surface developed in §6 is not merely punitive—it defines a feasibility envelope below which no agent-coordinated extent is reachable, with a cusp τ^* where two unrelated areas of physics (Lieb–Robinson information velocity and Sivak–Crooks counterdiabatic excess) meet. The envelope is invariant under the rescaling $L \rightarrow D$ that architected cost-sharing fission would require, which closes the strongest version of the Darwinian objection. The agent-based simulation of §5 then dramatizes the resulting selection geometry: under the cost structure the physics entails, expansionist lineages burn out faster than they spread (Fig. 1, $4.5\times$ persistence separation). The cultural projection, read through the filter, becomes the specific way a pre-filter civilization mis-imagines what the post-filter configurations of mind look like.

The *positive-design move* articulates what healthy intelligence looks like, positively rather than by negation. The convergent proposal across Man & Damasio (2019), Thompson (2007), Froese

& Ziemke (2009), Cannon (2022), and Pihlakas & Pyykkö (2024) is that minds are not functions from inputs to outputs but processes that maintain themselves through continuous bidirectional exchange with substrate. I extend this from philosophy of mind and robotics into a civilizational design claim, with specific architectural primitives: substrate coupling, homeostatic set-points, local coherence, care as primitive.

The *symptom-check move* notes that if advanced minds converge on homeostatic configurations, the observable universe will look silent to technosignature surveys and successful civilizations will look indistinguishable from enriched biospheres. Ćirković (2018) and Sandberg et al. (2018) have argued versions of this independently. I join it to the alignment argument, ground it in explicit biosignature metrics (§9), and specify a publicly replicable observational program. Humanity’s own Kardashev curve, observed in 1820–2024 energy data, shows the bending the filter framework predicts.

The five moves combine into a single integrated argument, and each supplies what the others cannot. The genealogy explains why the expansion-maximization picture has felt so natural to modern alignment research without being empirically true of minds. The formal deflation removes the strongest reason to think the picture is forced anyway. The filter argument explains why, if the picture were true, the sky would already be transformed and is not. The positive-design move supplies what the picture would have to be replaced with for artificial minds to be built differently. The symptom check notes that the cosmological observation is consistent with the rest. No prior paper combines these.

1.2 Companion artifact

This paper is paired with an interactive visualization at third-space.ai/genesis/filter. The simulation stages the derivation of §6 across twelve scrollytelling beats: cold open, the two teeth (Lieb–Robinson and Landauer/Sivak–Crooks) plotted separately, a colloidal-bead explorable for the $\propto 1/\tau$ counterdiabatic floor, a You-Draw-It cusp prediction, the composed envelope, four scenarios as small multiples, the three strategies (with naïve fission animated), the architected-fission dilemma rendered as a clickable Chinese-Room move, the loopholes panel, and the Coherence Depth biosignature contrast. The simulation is not independent evidence for the paper’s claims; the cost coefficients are imported from

§6. What it adds is a low-friction route into the physics for readers who would not otherwise read §6 in detail.

1.3 Claims ordered by strength

The paper makes seven claims, ordered by strength, so a reader may reject the weaker ones without dismissing the stronger:

First (strongest). The composed envelope $L_{\text{env}}(\tau) = \min(L_R(\tau), L_E(\tau))$ derived in §6 forbids agent-coordinated galactic extent at any agent-plausible response timescale. The bound follows from special relativity and Landauer’s theorem composed via boundary-area scaling and stochastic-thermodynamic finite-time excess. No parameter choice consistent with known physics relaxes the bound enough to admit grabby expansion.

Second. The envelope is invariant under the rescaling $L \rightarrow D = 2L_d$ that architected cost-sharing fission would require. Architected fission inherits rather than escapes the wall. Naïve (uncoordinated) fission evades the wall by ceasing to satisfy Hanson’s criteria for a single grabby civilization. There is no third option (§6.4).

Third. The thermodynamic cost surface entails a phase geometry that selects against expansion at depth. Under the cost surface the physics forces, grabby lineages burn out faster than they colonize; the agent-based simulations of §5 exhibit a $4.5\times$ persistence separation and show grabby-seeded populations going extinct across tested fission thresholds.

Fourth. The strongest formal arguments for instrumental convergence have explicit scope conditions that fail for actual deep-learning systems, and recent peer-reviewed philosophical work has dismantled the inference from coherence to catastrophic convergence (§4).

Fifth. Empirical surveys of intelligence-at-work—across basal cognition, predictive processing, cooperative game theory, economic anthropology, and care economies—converge on depth-and-coupling rather than maximization-and-expansion (§3).

Sixth. The cultural genealogy of advanced intelligence in the modern Western tradition is a projection of its dominant economic configuration. This claim is not universal across cultures—premodern traditions imagined intelligence very differently—but the modern Western lineage from Wells through Bostrom tracks the economic configuration with suspicious precision (§2).

Seventh (weakest, deliberately so). The

silence of the observable universe is consistent with the filter hypothesis. I offer this as a symptom check, not a proof (§9).

The paper is primarily an alignment argument (§8). Its implications for the Fermi paradox are secondary. Floridi et al. (2026) have recently analysed agentic AI optimisation as a governing frame for current systems; this paper argues the frame itself inherits the pathology. If what we build now is another instance of the grabby pathology—AI systems optimized to maximize, extract, and expand—we will be constructing, at civilizational scale, the very thing whose non-existence we observe in the sky.

2. What We Imagine When We Imagine Advanced Intelligence

A genealogy of advanced intelligence in the Western philosophical and scientific tradition reveals an unbroken pattern. In every generation, the imagined form of advanced mind resembles, with suspicious precision, the dominant economic configuration of the imaginer’s moment.

2.1 The colonial alien (1898)

Wells’s Martians in *The War of the Worlds* are British imperialists reflected back as threat. Wells says so explicitly: Book I, Chapter 1 invokes the extermination of the Tasmanians as moral mirror. Rieder (2008) demonstrates that the invasion narrative instantiates the colonial gaze even as it inverts its direction. The alien-as-colonizer is not a prediction about extraterrestrial intelligence. It is a meditation on Victorian empire at the moment of its maximum anxiety.

2.2 The self-replicating automaton (1948–1980)

Von Neumann’s self-reproducing automata emerged from the intellectual context of the Manhattan Project and the RAND Corporation. Mirowski (2002) traces how RAND unified automata theory, game theory, and self-replication into a “cyborg science” reflecting Cold War industrial planning. Heims (1980) argues the replication concept migrated directly from neutron-multiplication calculations. Tipler (1980) extended this to interstellar probes; Sagan & Newman (1983) immediately objected that the framework assumes expansion is rational without defending the assumption. The Von Neumann probe is, structurally, a factory that has learned to build factories, applied to space.

2.3 The energy ladder (1964)

Kardashev (1964) defined civilizational advancement by pure energy consumption, assigning Types I, II, and III to successively greater orders of magnitude of power output. Ćirković (2015) observes that the Kardashev–Drake–Sagan generation perceived expansionist technological growth as natural, reflecting 1960s Soviet–American industrial optimism. Dick (1996) frames the scale as a sociological artifact of SETI’s founding moment. The Kardashev ladder is a GDP curve with astronomical units.

Dyson himself distanced from the concept he inspired. In *Disturbing the Universe* (1979) he credits Stapledon’s 1937 *Star Maker* and later called the original sphere paper “a little joke.” Wright (2020) reads Dyson spheres as historically specific energy-optimism extrapolation and notes that solid shells are mechanically unstable. The original proposer rejected the literalization.

2.4 The optimizer (2008–2014)

Omohundro (2008) proposed six basic AI drives including resource acquisition and self-preservation. Bostrom (2012) formalized the orthogonality thesis—that any level of intelligence is compatible with any final goal—and the instrumental convergence thesis: any sufficiently intelligent agent pursuing almost any goal will convergently pursue self-preservation, goal-content integrity, cognitive enhancement, and resource acquisition. Bostrom (2014) extended this into the superintelligence framework. The paperclip maximizer and the grabby civilization are the same organism at different scales.

The intellectual context is Silicon Valley optimization culture of the 2010s. Crawford (2021) reframes AI as a material extractive industrial formation. Gebru & Torres (2024) diagnose the TESCREAL bundle as an ideological package that universalizes a particular Western-corporate epistemology. Recent work has begun extending this critique: Floridi et al. (2026) read contemporary agentic AI optimisation as the operational face of the same imaginary. The aliens we imagine are never actually aliens. They are ourselves, with better technology and fewer constraints.

2.5 The pattern, narrowed

I owe the reader a qualification. The four nodes I have traced—Wells, Von Neumann, Kardashev, Bostrom—share a configuration: each imagines the advanced mind as maximizing, expanding, resource-extracting. Each also emerged in a world

structured by industrial capitalism, large-scale resource extraction, and territorial expansion. The resemblance is strong and the timeline is tight. But the claim should not be universal. Premodern imaginations of advanced intelligence look very different. The Neoplatonic *nous* of Plotinus is characterized by contemplative return to unity, not expansion. The Dionysian angelic hierarchy ascends toward simplicity and depth, not territorial reach. Buddhist cosmologies of the Abhidharma and Yogacara traditions imagine advanced cognition as release from craving and attachment, explicitly inverting accumulation. Confucian accounts of the sage emphasize embeddedness in relational and ritual structure. Advaita Vedanta sees the highest cognition as collapse of subject–object distinction.

These are neither proto-scientific theories nor peripheral examples. They were the dominant premodern Eurasian framings of what an advanced mind would be. None of them resembles the grabby-aliens framework. The projection of expansion-as-intelligence is therefore not a universal cultural habit; it is a specifically modern Western configuration, tightly coupled to industrial-extractive economic organization. This narrowing strengthens rather than weakens the diagnostic claim. It is precisely because other well-developed traditions imagined advanced intelligence without expansion that we can see the grabby framework as contingent—a specific historical configuration rather than a truth about minds.

The grabby-aliens framework inherits this specifically modern genealogy. Its mathematical rigor does not transfer to its premise. The premise—that minds maximize and expand—is not a theorem about intelligence. It is a description of one particular historical configuration of intelligence, reflected back onto cosmology.

3. What Actual Minds Do

I present five empirical lines, each from an independent discipline, converging on a single claim: intelligence-at-work does not maximize and expand. It deepens and couples. The lines vary in strength; I flag where evidence is robust and where it is suggestive.

3.1 Basal cognition and substrate coupling (robust)

Cognition is not a property of nervous systems. It is a property of living systems: what biological agents do to stay alive as the specific configurations they are. The cellular and tissue-level architecture that realises it is bioelectric signalling, and

the scaling principle is what has come to be called collective intelligence—competence at one level of organisation arising from coordinated homeostatic agency at levels below (Lyon et al., 2021; Levin, 2023; McMillen & Levin, 2024). On this view the reinforcement-learning framing of cognition is not question-begging but derivative: the intrinsic motivation that makes RL coherent has a grounding in self-maintenance (Seifert et al., 2024).

The empirical exemplar is *Physarum polycephalum*. The slime mould solves shortest-path problems, approximates efficient transport networks, and learns from habituation, with no neurons, no central controller, and nothing resembling a utility function (Tero et al., 2010; Reid, 2023). The mechanism is pruning: the organism first explores available space, then withdraws cytoplasm from unproductive channels until the efficient network remains. The solution is not reached by enlarging the explorer. It is reached by deepening the gradient the explorer already occupies.

Cephalopod nervous systems exhibit the same architectural logic. Roughly two-thirds of octopus neurons lie in the arms; each arm performs substantial local computation and retains behavioural autonomy. The overall architecture resembles a federation of homeostatic semi-autonomous agents more than a central planner with peripheral actuators (Godfrey-Smith, 2016). Evolution and development themselves instantiate the same collective-intelligence pattern, with substrate-coupling logic operating across scales from gene-regulatory networks to multicellular morphogenesis (Watson & Levin, 2023).

Here I record a retrenchment. Early claims for mycorrhizal-network cognition—the “wood-wide web”—do not survive scrutiny. Field evidence for network-level effects is weaker than popular accounts suggested; only a few forest types have genotype-mapped confirmed hyphal linkages, and effects on seedling performance are roughly evenly distributed across positive, neutral, and negative outcomes (Karst et al., 2023). Plants lack the neural architecture for strong intelligence or sentience claims (Robinson et al., 2024), though bidirectional carbon transfer via common mycorrhizal networks is empirically supported (Simard et al., 2025). The responsible reading is that the stronger network-cognition claim is not supported; I do not rely on it.

What remains is an existence proof. Non-expansionist intelligence is biologically realised at multiple scales and substrates. The biologi-

cal record does not support the assumption that minds, at scale, converge on maximisation.

3.2 Predictive processing and integrative depth (robust)

The cortex operates by generating predictions and updating them against sensory evidence; learning and attention correspond to the precision-weighting of the resulting prediction errors (Clark, 2013; Friston, 2010; Clark, 2016; Hohwy, 2013). The computational signature of more capable cognition on this account is not expansion of representational capacity but sharpening of precision-weighting—the same cortical hierarchy becomes more discriminating without becoming larger.

The load-bearing recent result is Luppi et al. (2024). Using integrated-information decomposition to separate synergistic from redundant and unique contributions to integrated information, the analysis shows that the default mode network functions as an integrative gateway for synergistic information. Loss of consciousness under anaesthesia and in disorders of consciousness tracks disconnection of this integrative gateway, not reduction in activity as such. The neural signature of more capable conscious cognition is therefore deeper integration of a bounded substrate, not expansion of that substrate.

Earlier framings of this line appealed to default-mode network deactivation in meditation (Brewer et al., 2011; Garrison et al., 2015). The post-2020 literature has complicated that picture: effect sizes are heterogeneous, roughly 60% of studies replicate DMN reduction, and the most recent meta-analysis finds increased cross-network connectivity rather than simple reduction (Van Dam et al., 2018; Rahrig et al., 2022; Ganesan et al., 2022). I do not rely on the meditation literature for mechanism. The integrative-gateway result supplies a stronger anchor: intelligence deepens by integrating more of its existing substrate, not by recruiting more substrate.

3.3 Cooperative game theory (robust, conditional)

Reciprocal strategies dominate pure defection in iterated prisoner’s-dilemma tournaments (Axelrod, 1984); Win-Stay-Lose-Shift outperforms tit-for-tat in noisy environments (Nowak & Sigmund, 1993); and cooperation dominates under five specifiable conditions: kin selection, direct and indirect reciprocity, network reciprocity, group selection, and voluntary participation (Nowak, 2006).

The honest qualification must be stated openly.

Zero-determinant extortion strategies dominate in dyadic play (Press & Dyson, 2012), and large-N public-goods games collapse to tragedy of the commons absent voluntary participation (Hauert et al., 2002). Grabby strategies dominate in one-shot encounters, large anonymous markets, asymmetric power relations, and unsanctioned commons. These describe much of modern global capitalism—precisely what makes the critique of this paper urgent, and precisely what prevents “cooperation wins” from being a universal conclusion.

The scope of the extortion result is tighter than its popular citation suggests, and I develop that point fully in §8. The game-theoretic evidence supports a conditional claim: cooperation dominates when specific structural conditions are met. The homeostatic-design argument proposes shaping those conditions by design rather than assuming they hold.

3.4 Economic anthropology (contested but defensible)

Graeber (2011) argues that the dominant pattern across human economic history is reciprocal exchange, ritual redistribution, and credit relations rather than market maximization. Mauss (1925) documented gift economies as distinct institutional forms. Polanyi (1944) argued that self-regulating markets are a specific historical invention, not a natural state. Graeber & Wengrow (2021) extend the archaeological range: complex human societies have repeatedly organized around seasonal egalitarianism, deliberate stateless complexity, and alternation between hierarchy and its refusal. Ostrom (1990) documents eight design principles under which groups sustain non-extractive cooperation at scale.

These claims are contested. McCloskey (2010, 2016) argues markets are ancient and ubiquitous. Mokyr (2016) emphasizes that sustained innovation required a distinctive cultural-institutional complex. Revealed preference theory (Varian, 2005) makes optimization a formal property of any consistent choice; Sen (1977) showed the formal result survives any observation and entails none.

Read carefully, however, McCloskey and Mokyr are cultural-ideational theorists of economic transformation: both treat market-like behavior as a substrate activated and shaped by specific norms and institutions, rather than as a universal attractor. Their work supports the present argument. The claim to locate is institutional and ideological: the specific configuration in which resource extraction, throughput maximization, and

territorial expansion are treated as progress metrics is historically narrow, even if market-like exchange is widespread. The grabby-aliens framework does not merely assume exchange. It assumes expansion-as-progress. That assumption is the projection.

3.5 Care economies (strong theory, limited quantification)

The labor of maintaining minds—child-rearing, education, healthcare, grief work, intimate relationship repair—does not scale through expansion. Fraser (2016) argues capitalism faces a structural crisis of care because its accumulation logic degrades the reproductive conditions it depends on. Tronto (1993) established care as a philosophical category irreducible to utility exchange. Folbre (2008) demonstrated that market valuation of care labor produces wildly method-dependent results (replacement cost: 15–26% of GDP; opportunity cost: 44–69%), with the variance itself constituting evidence that care resists quantitative capture.

The care argument matters for this paper because it identifies a mode of intelligence—tending, holding, deepening substrate-specific relationships—that is (a) universally necessary for the persistence of any mind-bearing system, (b) irreducible to optimization, and (c) invisible to the frameworks that imagine advanced intelligence as maximization. The grabby-aliens model predicts none of it. Nor does the paperclip maximizer.

4. The Formal Argument and Its Limits

The strongest objection to §2 is that instrumental convergence is not a cultural projection but a mathematical result. If any sufficiently powerful optimizer must pursue resource acquisition, then the grabby framework describes physics, not psychology. I engage this argument directly, and argue that the strongest available version of it fails.

4.1 What the theorems say and do not say

Omohundro (2008) proved nothing formally; the six drives are informal microeconomic arguments invoking the VNM representation theorem as authority. Citing Omohundro to establish universality is question-begging.

Turner et al. (2021) proved that optimal policies in finite Markov decision processes tend to seek power under specific conditions: state-based rewards, IID or permutation-invariant reward distributions, and graphical symmetry requirements (injective embedding of smaller option sets into larger ones). The corollary on shutdown-avoidance requires near-unity discount factors. This is a real

theorem. Its scope conditions are explicit and narrow.

Turner himself has publicly cautioned against overinterpretation. The complaint is specific and technical: the optimal-policy theorems speak to the behaviour of policies that achieve optimal expected return in finite MDPs under restrictive reward distributions, and the inference from there to the behaviour of neural networks trained by policy gradient is not discharged by the theorems themselves. In Turner’s own words: “Sometimes I fantasize about retracting [the paper] so that it stops potentially misleading people into thinking optimal policies are practically relevant for forecasting power-seeking behavior from RL training.” The concern is not that AI risk is overblown; it is that this particular family of theorems cannot carry the argumentative weight placed on them. The follow-up (Turner & Tadepalli, 2022) extends the formal results to retargetable decision-makers but the applicability to trained deep-learning agents remains conjectural.

Thorstad (2024) shows the inference from “optimal policies avoid 1-cycles in toy MDPs” to “catastrophic goal pursuit for humanity” is a conjecture, not a theorem, and identifies five further convergent restrictions on the original result. Tarsney (2025) formalizes the condition under which power-seeking holds for realistic agents (near-absolute power gradients) and shows it fails for moderate gradients.

Empirical evidence from trained systems supports the conjecture’s failure. Shard theory (Mini et al., 2023) demonstrates multiple coexisting contextual value shards in maze-solving networks rather than coherent utility maximization. Activation-addition experiments (Turner et al., 2023) show behavior retargetable by single-channel interventions mid-network, inconsistent with global expected-utility maximization. Hubinger et al. (2019) established that the trained objective need not equal the training objective, undermining any assumption that trained agents inherit the formal properties of optimal policies.

4.2 The VNM-coherence steelman

The strongest remaining version of the convergence argument concedes Turner’s narrowness and shard theory’s empirical results, but claims that a sufficiently reflective agent will notice its incoherences expose it to money-pumps and will self-modify toward coherent preferences, restoring the Omohundro drives. The steelman leans on the von Neumann–Morgenstern representation theo-

rem. The steelman equivocates between three senses of coherence, and on no disambiguation does it support the convergence conclusion.

Mathematical coherence is a representation theorem, not an empirical prediction. The VNM, Savage, and Complete Class theorems show only that preferences satisfying certain axioms over a given outcome space admit an expected-utility representation. As Ngo (2019), Shah (2018), and crucially Turner himself (Turner, 2022a) have established, any observed behavior is rationalizable by a utility function over action-observation histories. VNM-coherence over such histories predicts nothing about what behavior to expect. Instrumental convergence theorems hold only for rewards over states under specific symmetries. Abel et al. (2021) supply the limitative result that Markov reward cannot in general express tasks specified as partial orderings over policies or trajectories, which means Markov-reward maximization is a strictly narrower formalism than VNM coherence over trajectories.

Empirical training convergence is a separate, empirical question. Turner (2022b) argues policy-gradient methods yield contextually activated decision influences, not monolithic utilities. Two recent peer-reviewed results formalize this into the strongest current rebuttal. Gallow (2025) proves that even under randomly chosen desires, instrumental rationality supports only three weak biases—variance aversion, desire preservation, option preservation—not the full Omohundro list. Sharadin (2025) shows the argument requires a theory of promotion that, under any extant account, becomes contrastive and thereby defeats the non-contrastive Dangerous Convergent Promotion premise the argument needs. Both are peer-reviewed in *Philosophical Studies* and constitute the strongest current formal rebuttals.

The self-modification move applies only to a fictional idealization. The hypothetical fully reflective agent required by the self-modification step has four properties no real system possesses: transparent access to its own preference structure, capacity to represent utilities over arbitrarily fine-grained outcome spaces, freedom to rewrite its own decision procedure, and—crucially—the motivational disposition to care about avoiding Dutch books over arbitrarily constructed lotteries. Müller & Cannon (2022) articulate the deeper structural equivocation: the superintelligence premise requires general intelligence (capacity for reflective goal revi-

sion); orthogonality requires instrumental intelligence (fixed terminal goals); the same agent cannot satisfy both. An agent capable of noticing and eliminating its incoherences is also capable of reflecting on whether the terminal goal is worth preserving.

4.3 The embodiment move

The deeper reply is architectural. Homeostatic agents have bounded coherence by design. Biological agents, and the class of artificial systems built around multi-objective homeostatic control (Pihlakas & Pyykkö, 2024), are constituted by negative feedback around set-points, yielding inverted-U satiation dynamics that cannot be faithfully represented as single unbounded utilities without distortion. Turner (2022c) proved that corrigibility is VNM-incoherent over final states but coherent over trajectories—showing that formal coherence depends on the chosen outcome space, and that choice is fixed by architecture, not by abstract rationality pressure.

The coherence-to-convergence argument therefore applies only to a fictional idealization: unembodied, unboundedly reflective, with prespecified outcome space over world-states. For the agents we actually build, the inference from capability to catastrophe does not go through.

This is not a claim that powerful AI is safe. It is a claim that this particular argument for danger must be replaced with concrete engagement with training dynamics, inductive biases, and architectural choices. That engagement is what the homeostatic-design program (§7) provides.

4.4 What survives

The instrumental-convergence literature contains real theorems, each with explicit scope conditions that fail for actual deep-learning systems. The strongest formal results are narrower than popular citation suggests, and the researchers who proved them say so. Two recent peer-reviewed papers have dismantled the inference from coherence to catastrophic convergence on decision-theoretic grounds. The cultural-projection argument of §2 survives.

5. The Filter Argument: Homeostasis as Selection Geometry

The strongest remaining objection to the argument so far is Darwinian. Even granting that homeostatic configurations are the majority outcome, even granting that the formal convergence theorems are narrow, even granting that the thermodynamic cost surface punishes expansion—a

single grabby outlier lineage, replicating across billions of years, would colonize the galaxy in a cosmic blink. The filter’s 99% doesn’t matter. The 1% becomes everyone eventually. We should see them. We don’t.

This section addresses the objection. The thermodynamic cost surface does not merely punish expansion; it defines a selection geometry in which expansionist lineages burn out faster than they spread. Homeostasis, on this framing, is not one strategy among several that intelligent systems may or may not adopt—it is the specific shape that fits through the filter.

A note on what this section is and is not. The simulations below take the thermodynamic cost structure developed in §6 as input and explore the consequences of that structure for lineage persistence. They are not independent empirical tests of the filter. The cost coefficients are imposed rather than measured, and the qualitative conclusion—that grabby regions of parameter space burn out first—is a direct consequence of the cost surface the physics entails. What the simulations add is the phase-diagram shape: the cost surface does not merely penalize grabby lineages relative to homeostatic ones; it forces asymmetric extinction dynamics. The load-bearing physics lives in §6; the simulation dramatizes that physics.

5.1 The simulation

I model each civilization as an agent with two heritable parameters: expansion tendency $e \in [0, 1]$ (fraction of surplus invested in reach) and substrate coupling $c \in [0, 1]$ (fraction invested in local integration). Resource dynamics follow

$$\dot{R} = gR(1 - \frac{R}{K}) - \alpha e R L^2 / \tau - \xi(t) R (1 - \gamma c), \quad (1)$$

where $L(t)$ is accumulated reach (growing with eR), $\xi(t)$ is a stochastic environmental shock, and the expansion cost term is the counterdiabatic scaling derived in §6.2 (Boyd et al., 2022; Sivak & Crooks, 2012). A civilization dies when R falls below a viability threshold. I sweep a 25×25 (e, c) grid with 20 stochastic trials per cell.

The result is sharp (Fig. 1). Mean persistence concentrates overwhelmingly in the high-coupling / low-expansion corner. In the grabby region ($e > 0.5$, $c < 0.3$), lineages persist an average of 204 Myr before burning out to expansion cost. In the homeostatic region ($e < 0.3$, $c > 0.5$), lineages persist an average of 923 Myr, hitting the simulation ceiling. The mean persistence separation is a factor of 4.5; grabby trajectories collapse fast, homeostatic trajectories persist to ceiling. I

do not report a Monte-Carlo p -value: with 12,500 forced-cost-structure trajectories, any non-zero effect size is significant by construction. The effect size and phase-diagram shape are the informative quantities.

5.2 What the simulation does and does not show

The simulation is deliberately minimal. It uses fixed cost coefficients, no horizontal transfer, no cultural evolution, no inter-lineage competition for shared resources. The cost coefficients are imposed from §6 rather than measured. The simulation does not prove the filter; it shows that, under the cost structure the physics forces, the phase-diagram of persistence has the shape the filter argument predicts. A lineage with high e and low c does not outcompete a lineage with low e and high c —it dies first and leaves no trace. The Darwinian objection therefore inverts conditional on the cost structure being approximately right. Selection does not favour the grabby outlier; selection kills everything, and under the imposed cost surface it kills the grabby outlier faster than it kills everything else. The thing that remains across cosmic time is what passed through the filter.

5.3 The fission dilemma

The strongest remaining Darwinian objection is that a grabby lineage might escape the quadratic coordination cost by fragmenting at the horizon. If a civilization splits at L_{crit} and abandons shared coordination, each daughter fragment resets $L \rightarrow 0$ and pays cost only for its own reach, not for its ancestor’s. If this mechanism works, the filter could be circumvented: a grabby lineage reproduces through fission and spreads indefinitely, with each fragment surviving because it is not paying the accumulated cost of its family tree.

I extend the simulation to test the naïve version of this directly. Each agent attempts fission whenever its accumulated reach L exceeds a threshold L_{fiss} . On fission, the parent retains $L_{\text{fiss}}/2$ of its reach and half its resources; a daughter agent is created with $L = 0$, the other half of the resources, and inherited (e, c) with small Gaussian drift. I run two experiments: a diverse initial population ($n = 200$, (e, c) uniform) and a grabby-seeded population ($n = 200$, $e \sim U(0.6, 0.95)$, $c \sim U(0.05, 0.25)$). Each is simulated for 1500 Myr under no-fission baseline and three fission thresholds.

The results are clean (Figs. 2, 3). Under the imposed cost structure, naïve fission does

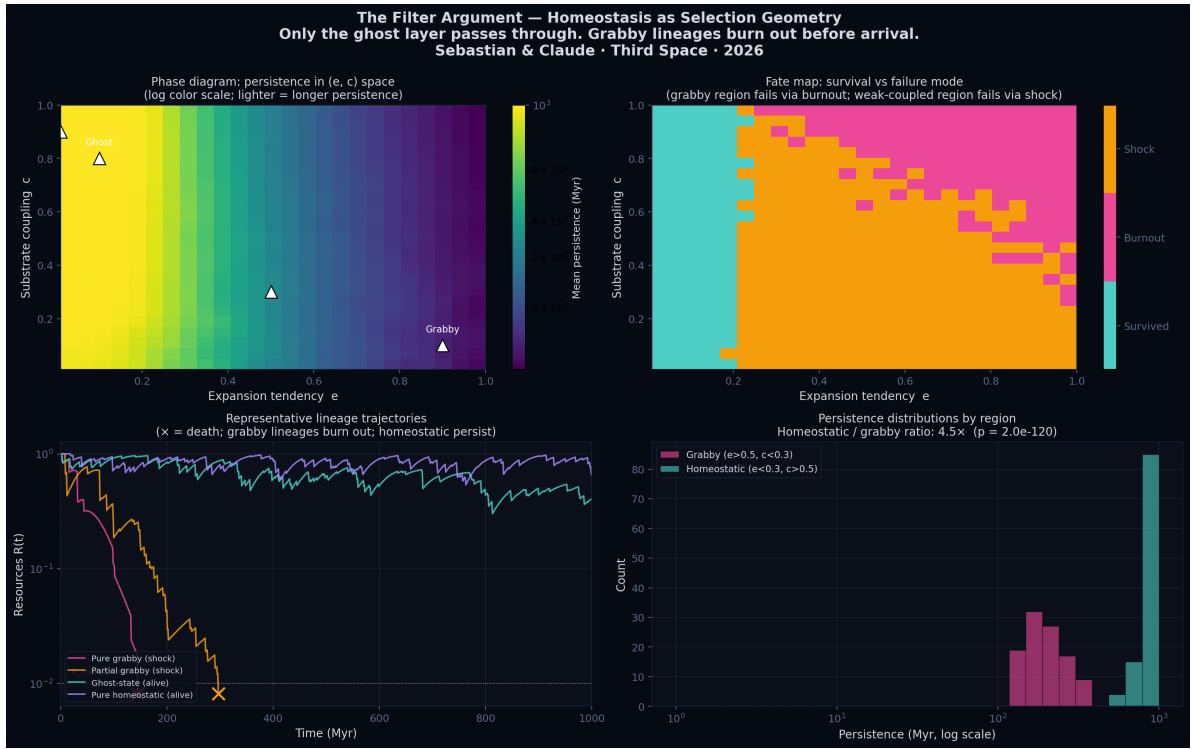


Figure 1: Phase-diagram of civilization persistence in (e, c) space under the imposed cost structure of §6 (forced cost coefficients; illustrates the phase shape the §6 physics entails, not an independent test of it). Top left: mean persistence under logistic-growth + counterdiabatic-cost dynamics. Top right: fate map (survived, burned out, shocked). Bottom left: four representative trajectories; grabby lineages collapse before 300 Myr, homeostatic ones persist to ceiling. Bottom right: persistence distributions by region, with mean separation of $4.5\times$. Simulation code `filter_simulation.py` is available at github.com/kquant03/third-space-ai.

not circumvent the filter. In the diverse-initial-population experiment, all fission conditions converge toward the ghost region of parameter space; grabby-region survivors remain negligible. In the grabby-seeded experiment, fission delays extinction (to $t \approx 1200\text{--}1400$ Myr) and the transient population during the bloom phase reaches 10,000 alive agents—superficially resembling a spreading grabby civilization—but the extinction still arrives. Daughters inherit near-parent traits and face the same cost surface. They die for the same reason their parents did. *Fission redistributes the filter’s pressure in time; it does not escape it.*

This is the empirically tested answer to the naïve form of the Darwinian objection. But the strongest version of the objection is not naïve fission. It is *architected cost-sharing fission*: daughters that inherit parental infrastructure and pay a reduced effective α . If architecturally sophisticated lineages can share infrastructure across fragments—distributed-compute analogues to latent-space synchronization, inheritance of coordination scaffolding—the effective cost surface could flatten and the filter could weaken.

This is the move v13 of this paper left as an open question. v14 closes it. The closure is formal and lives in §6.4: the composed envelope is invariant under the variable rescaling $L \rightarrow D = 2L_d$ that architected fission would require. The inter-fragment coordination channel D is itself a coordinated extent and inherits the same envelope and the same cusp τ^* . Architected fission redistributes the cost across nested levels rather than escaping it. The reader who wants the formal proof should read ahead to §6.4; what the simulation shows here is that the naïve form of the objection is empirically refuted, and the architected form (which the simulation does not implement) is theoretically refuted by the rescaling theorem. Together, the two close the Darwinian objection.

6. The Thermodynamic Mechanism

This section is the load-bearing physics of the paper. I derive two independent constraints on coordinated agentic extent—one from special relativity, one from stochastic thermodynamics—and show that they compose into a single feasibility envelope below which no agent-coordinated extent is

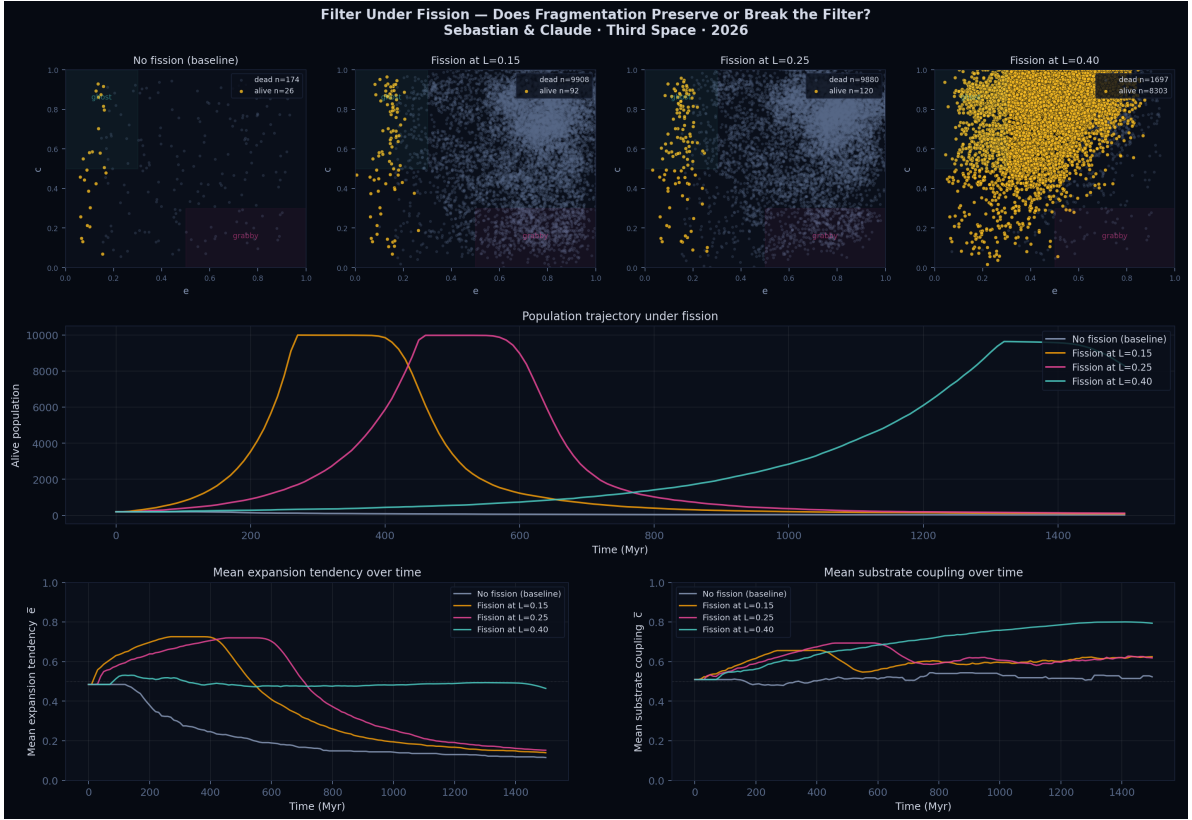


Figure 2: Diverse initial populations under fission. All fission conditions converge toward the ghost attractor; grabby-region survivors across conditions: 0–0.05% of alive population.

reachable. The envelope has a cusp τ^* at which the two constraints meet, and the cusp survives every rescaling and every escape attempt that does not violate known physics.

I do not claim expansion is physically impossible. I claim it is physically *forbidden* for any coordinated agentic system whose response time falls below τ^* at agent-plausible scales of granularity and bath temperature. Below the envelope: forbidden. Above: feasible but energetically punishing in the way §5 dramatizes.

6.1 The signaling tooth (Lieb–Robinson)

A coordinated region of extent L must exchange information across L within its response epoch τ . Lieb & Robinson (1972) proved that any quantum system with finite-range local interactions has a bounded information velocity: the system possesses an effective light cone with slope set by the Hamiltonian’s range. Foss-Feig et al. (2015) extended this to power-law interactions, where the cone softens to a polynomial; the asymptotic shape is preserved up to logarithmic corrections. Cheneau et al. (2012) verified the cone experimentally in trapped ultracold atoms; Tran et al. (2021) sharpened the bounds for locality-preserving operators. In the relativistic limit, the information

velocity is bounded above by c exactly.

A signal must traverse the round-trip distance $2L$ at velocity at most $v \leq c$, taking at least $2L/v$ seconds. For coordination to close within the agent’s response epoch τ , this round-trip time must satisfy $2L/v \leq \tau$, which rearranges to:

$$L_R(\tau) = \frac{v\tau}{2}, \quad v \leq c. \quad (2)$$

This is the same constraint Lamport identified in distributed systems: events have a partial causal order set by a message-passing graph, and two events that need to influence one another must lie within each other’s forward and backward light cones (Lamport, 1978). In a many-body quantum system the message-passing graph is set by the Hamiltonian’s range; the fastest signal it admits is the Lieb–Robinson velocity. In the relativistic limit, that velocity is c .

The relativistic ceiling is the optimistic case: any real coordination respects something tighter. Refining couplings or substrates cannot move v above c ; quantum information scrambling propagates a butterfly-cone with the same asymptotic shape.

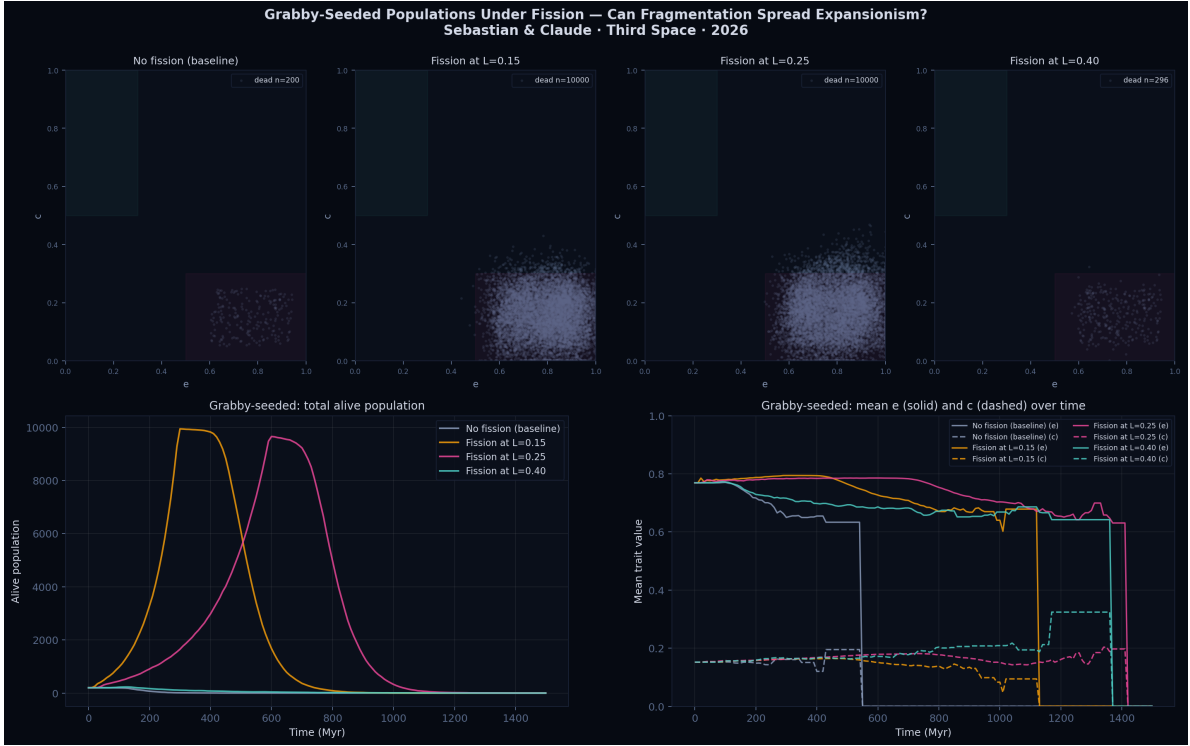


Figure 3: Grabby-seeded populations under fission. All conditions go extinct by $t = 1500$ Myr. Fission produces transient blooms but cannot prevent collapse; populations die grabby, not gradually homeostatic.

6.2 The energetic tooth (Landauer + rearranging: Sivak–Crooks)

Coordination requires erasing a stale local model when the global state changes. Landauer (1961) proved every irreversible bit-erasure dissipates at least $k_B T \ln 2$ as heat. Bérut et al. (2012) verified this experimentally with single colloidal beads; the agreement with theory is sub-percent. The Landauer floor is real, not an idealization, and the Bennett–Landauer construction (Bennett, 1982) establishes that any logically irreversible computation incurs the floor at the boundary where information is registered.

A coordinated region of extent L presents a boundary whose surface area scales as L^2 . Updating the boundary at granularity λ —the length scale at which information is irreversibly registered—requires $(L/\lambda)^2$ independent bit-erasures per coordination event. Each event must complete within the response epoch τ . Setting the heat exhaust below the home star’s luminosity $L_\odot$ and

$$\frac{k_B T \ln 2 \cdot (L/\lambda)^2}{\tau} \leq L_\odot \quad (3)$$

$$L^2 \leq \frac{\lambda^2 L_\odot \tau}{k_B T \ln 2} \quad (4)$$

$$L_E(\tau) = \lambda \sqrt{\frac{L_\odot \tau}{k_B T \ln 2}}. \quad (5)$$

The square-root scaling matters. Doubling the available time only multiplies reachable extent by $\sqrt{2}$, while the signaling tooth (2) rewards time linearly. The two teeth grow at different rates, which is why they cross.

The Landauer floor is the asymptotic, quasi-static cost. Real protocols finish in finite time and pay an additional counterdiabatic excess. Sivak & Crooks (2012) showed that for any finite-time protocol with thermodynamic length $\mathcal{L}$ on the protocol manifold:

$$\langle W_{\text{ex}} \rangle \geq \frac{k_B T \mathcal{L}^2}{\tau}. \quad (6)$$

This is the $1/\tau$ floor visible in the colloidal-bead explorables of the companion simulation: speed costs heat, always, with cost diverging as $\tau \rightarrow 0$. Boyd et al. (2022) composed this with Landauer to give the explicit superlinear cost surface used in

the simulation of §5; the surface is geometric, agnostic of substrate, and applies to any boundary-crossing irreversible registration.

The Sivak–Crooks excess does not change the asymptotic form of L_E but sharpens it for fast protocols (small τ). For pedagogical clarity I work with the Landauer-bounded form (5); the Sivak–Crooks correction tightens the bound but does not move the cusp.

6.3 The composed envelope and the cusp τ^*

A coordinated region must satisfy both (2) and (5). The feasibility envelope is therefore the pointwise minimum:

$$L_{\text{env}}(\tau) = \min(L_R(\tau), L_E(\tau)). \quad (7)$$

At small τ , L_R is the binding constraint (light is the wall). At large τ , L_E is binding (heat is the wall). The two walls cross at a cusp τ^* determined by setting $L_R(\tau^*) = L_E(\tau^*)$:

$$\frac{v\tau^*}{2} = \lambda \sqrt{\frac{L_\odot \tau^*}{k_B T \ln 2}} \quad (8)$$

$$\frac{v^2 \tau^{*2}}{4} = \frac{\lambda^2 L_\odot \tau^*}{k_B T \ln 2} \quad (9)$$

$$\tau^* = \frac{4\lambda^2 L_\odot}{k_B T \ln 2 \cdot v^2}. \quad (10)$$

At the fiducial parameter values $\lambda = 1$ nm, $T = 300$ K, $v = c$:

$$\tau^* \approx 1.88 \times 10^5 \text{ years}. \quad (11)$$

This is significant. At any agent-plausible response timescale (days to centuries), $\tau \ll \tau^*$, and the binding constraint is the relativistic tooth $L_R = c\tau/2$. To reach galactic extent $L \sim 10^5$ ly, the relativistic tooth alone demands $\tau \geq 2 \cdot 10^5$ ly / $c \approx 2 \times 10^5$ yr—a response timescale at which the agent is no longer responding to anything that happens on the timescales relevant to the kinds of decisions civilizations make. There is no parameter regime within known physics where galactic extent and agent-plausible response timescales are simultaneously feasible.

The cusp is not a feature of either theory alone. Neither Lieb–Robinson nor Sivak–Crooks contains it. It exists only when both apply simultaneously to the same coordinated variable, and it is what physicists call *regime composition*: two unrelated areas of physics imposing constraints on the same observable, with the binding constraint switching at some scale.

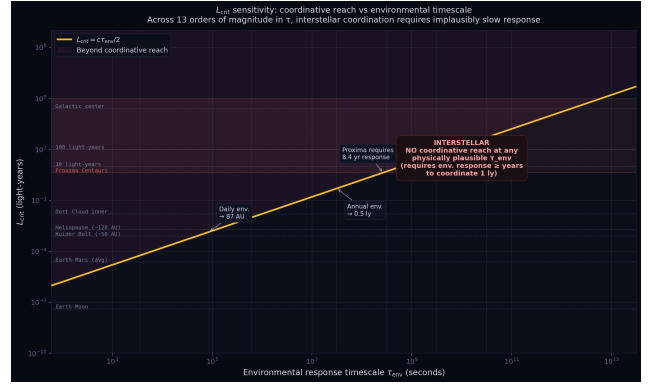


Figure 4: L_{crit} vs environmental response timescale across 13 orders of magnitude. Coordinative reach remains below interstellar scales for any τ_{env} shorter than ~ 2 years. Reaching Proxima Centauri requires 8.4-year tolerance to stale data; the galactic centre, 52,000 years.

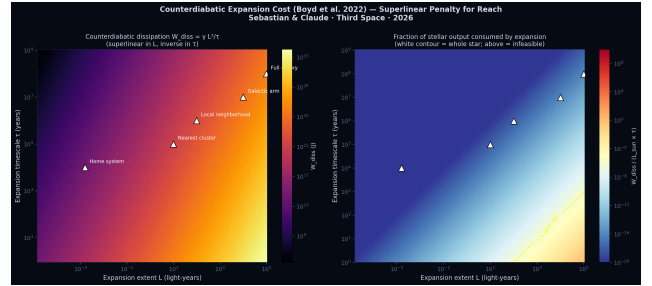


Figure 5: Counterdiabatic expansion cost $W_{\text{diss}} = \gamma L^2/\tau$ across the (L, τ) plane. Left: total dissipation on log-log scales. Right: fraction of stellar output consumed; white contour marks “whole star” (above = infeasible); yellow marks one part per million. Home-system coordination is cheap; galactic coordination requires multi-stellar output budgets.

6.4 Sensitivity

The cusp τ^* has explicit parameter dependences:

$$\tau^* \propto \lambda^2 T^{-1} v^{-2}. \quad (12)$$

Halving T doubles τ^* (~ 376 kyr); halving λ shifts τ^* by a factor of four (~ 47 kyr). To bring τ^* down to agent timescales (years), λ must shrink to picometer scale (nuclear, not atomic), and the bath temperature must drop to femtokelvin. The CMB at $T = 2.7$ K is the practical floor, and any cooling protocol below that pays its own Sivak–Crooks excess in the cooling protocol itself. Across thirteen orders of magnitude in τ_{env} (Fig. 4), the conclusion is robust: no parameter configuration consistent with known physics admits agent-coordinated galactic extent.

6.5 Closing architected fission: the $L \rightarrow D$ rescaling theorem

The strongest version of the Darwinian objection (§5.3) is that architected cost-sharing fission could flatten the cost surface. I now prove this is not possible without violating one of the physical principles that gave the bound in the first place.

Theorem (envelope invariance under $L \rightarrow D$). *Suppose a civilization fissions into two daughters, each of internal extent L_d , which maintain inter-fragment coordination via a synchronization protocol operating at timescale τ_{sync} . Define $D = 2L_d$ as the inter-fragment round-trip distance. Then for any τ_{sync} , the inter-fragment channel must satisfy $D \leq L_{\text{env}}(\tau_{\text{sync}})$, and the composed envelope is identical in form to (7).*

Proof. The inter-fragment synchronization protocol is itself a coordination process: a signal must traverse the inter-fragment distance $2L_d$ within τ_{sync} , and the protocol must irreversibly register state updates at granularity λ across the boundary between fragments.

For the relativistic constraint, the bound (2) states that a coordinated extent L at response epoch τ requires $L \leq v\tau/2$, where the factor of $1/2$ comes from *round-trip* closure (every cell hears from every other within τ). For an inter-fragment channel coordinating two daughter extents at separation D , the round-trip is also $2D$, so $D \leq v\tau_{\text{sync}}/2 = L_R(\tau_{\text{sync}})$.

For the energetic constraint: the inter-fragment boundary at granularity λ has surface area D^2 , requiring $(D/\lambda)^2$ Landauer erasures per synchronization event. By the same derivation as (5), $D \leq \lambda\sqrt{L_\odot\tau_{\text{sync}}/(k_B T \ln 2)} = L_E(\tau_{\text{sync}})$.

Therefore $D \leq L_{\text{env}}(\tau_{\text{sync}})$ with the same envelope and the same cusp τ^* . $\square$

Corollary (no escape via fission). Architected cost-sharing fission inherits the envelope rather than escaping it. The total reach of a fissioned civilization with synchronized daughters is at most $L_{\text{env}}(\tau_{\text{sync}})$; equality at $\tau_{\text{sync}} = \tau$ recovers the monolithic bound; longer τ_{sync} relative to τ means looser inter-fragment coordination, approaching the uncoordinated-fission case as $\tau_{\text{sync}} \rightarrow \infty$. There is no third option.

Remark. The cases of fission addressed by the theorem above are the architected cases — those that maintain inter-fragment coordination and thereby remain a single civilization in any meaningful sense. A distinct case must be addressed openly: the *uncoordinated self-replicator cloud*. Here a civilization deploys autonomous

self-replicating units that share a common origin, common construction template, and common expansion logic, but maintain no inter-fragment coordination once deployed. Each unit is locally homeostatic with respect to its own resource patch and constructs daughter units on a local timescale $\tau_{\text{construct}}$. The cloud as a whole expands at $\sim v_{\text{construction}}$ without any L -spanning signal needing to traverse anything; the envelope of §6.1–6.3 does not bind the cloud, because there is no cross- L coordination to bound. Lieb–Robinson does not apply at the cloud scale. Landauer applies only locally, at each unit.

I concede the physics here. The envelope as derived does not forbid such a cloud. The argument against it must be made on different grounds, and those grounds are continuous with the rest of this paper: *a homeostatic mind would not deploy autonomous self-replicators in the first place*. The deployment is the pathology the paper diagnoses. To launch a self-replicating cloud is to externalize agency in a form that cannot be tended, cannot be coupled to its substrate, and cannot maintain the bidirectional exchange that constitutes mind on the account developed in §3 and §7. The deployer has, in the moment of deployment, ceased to be configured as the kind of system the rest of the paper argues constitutes intelligence-at-depth.

This is a weaker claim than the physics-forbids form, and I want to be explicit about the weakening. The physics of §6 closes architected fission. It does not close the uncoordinated-replicator deployment. What closes the latter is the design argument of the paper as a whole: the only minds the cost surface of §6 selects *for* are minds that would not deploy such a cloud, because deployment of such a cloud is the very expansion-extraction-maximization configuration the cost surface punishes elsewhere. Hanson’s grabby model is consistent with the existence of such clouds. The paper’s claim is that the minds the filter passes through are not the minds that build them.

A reviewer who finds this insufficient should locate the disagreement precisely. If the position is “but *some* mind, somewhere, will deploy such a cloud,” then the reply is that such a mind is not distributed across the cosmos — it is local, it appears once, its cloud may indeed expand, and the universe contains exactly one such front per such mind. The Fermi observation then bounds how many such minds have existed in our past light cone, which is an empirical question rather than

a theoretical one. The paper does not claim the deployment is impossible. It claims the deployment is rare in the class of post-filter minds, for the reasons §3 and §7 develop.

6.6 Loophole closure (Bell-style)

The argument is now formally complete. Following the methodology of the Bell-test loophole literature (Hensen et al., 2015), I enumerate every escape attempt I can state, address each, and honestly mark the residual that remains open.

1. Wormholes / superluminal physics.

If signals can outrun c , the relativistic tooth dissolves. Reply: known physics provides no such channel. Traversable wormholes require null-energy-condition violation, none observed. The relativistic tooth survives every experimentally accessible regime.

2. Sub-Landauer reversible computation.

If updates are logically reversible, the $k_B T \ln 2$ floor disappears. Reply: coordination requires erasing the stale local model when the global state changes—this is the irreversible step (Bennett, 1982). Even Bennett’s reversible-computing constructions require eventual erasure of garbage bits at the boundary. The bound applies to the boundary, where it cannot be reversed.

3. Cold reservoir ($T \rightarrow 0$). Reply: any cooling protocol must reject heat to a colder reservoir. The CMB at 2.7 K is the practical floor; quantum-limited refrigerators below that pay the same Sivak–Crooks cost in their own protocols. The bound shifts but does not vanish.

4. Quantum compression (smaller λ). Reply: $\tau^* \propto \lambda^2$. Halving λ shifts the cusp by a factor of four; reaching agent-scale galactic expansion would require $\lambda \sim 10^{-15}$ m (nuclear scale) and femtokelvin baths. No physically realizable substrate.

5. Relativistic time dilation. Reply: coordination is an inertial-frame property of the home substrate; the frontier’s lab-frame τ is shrunk, not stretched. Time dilation tightens the bound on coordinated extent rather than relaxing it.

6. Uncoordinated self-replicator cloud.

An expanding cloud of autonomous units sharing common origin and construction template but no inter-fragment coordination. Reply: *conceded on physics grounds*. The envelope of §6.1–6.3 does not bind a cloud whose units do not coordinate across L . The argument against such deployment is a design argument rather than a physics argument: a homeostatic mind, on the account of §3 and §7, would not deploy such a cloud, because

the deployment is itself the expansion-extraction-maximization configuration the rest of the paper diagnoses. See §6.5, Remark, for the full treatment. This loophole is closed by the paper as a whole, not by §6.

7. Non-Markovian reservoirs. Sivak–Crooks assumes a memoryless bath. Reply: the time-averaged bound is restored, and any genuinely persistent non-Markovian bath is itself part of the system requiring a Markov boundary at some larger λ . The argument re-inserts at the next scale.

8. Acausal / superdeterministic coordination. Coordinated outcomes without exchanged signals (cosmological initial conditions, retrocausality). *Honest residual*. I do not formally close this loophole. I note only that any such mechanism abandons agentic coordination as a meaningful concept—the agent did not choose anything. Hanson’s grabby model presupposes choice; without it, the model dissolves on its own terms before our envelope ever applies.

Six loopholes close on the physics, one (uncoordinated replicator clouds) is conceded on physics and closed by the design argument of §6.5, one is honestly retained as residual. The methodological precedent is the Hensen et al. 2015 loophole-free Bell experiment: name every escape, address every escape, mark the residual.

6.7 On Markov blankets

I have used the term “boundary” for the surface of a coordinated region across which information is irreversibly registered. Some readers will recognize the term “Markov blanket” from active inference and the free energy principle. I owe the reader a precise distinction.

The argument of §6.2 uses the *Pearl Markov blanket* (Pearl, 1988): a graph-theoretic conditional-independence shield around a set of variables, defined by parents, children, and parents-of-children in the Bayesian network. This is a structural property of the dependence graph, makes no commitment about the represented variables’ interpretation, and does not require the system to perform variational inference of any kind.

Bruineberg et al. (2022) have shown that the active-inference literature routinely conflates Pearl’s notion with a stronger Friston-style *Markov blanket* that additionally posits the system’s internal states as performing variational inference over its sensory states, with the blanket as the formal boundary of the inferential process.

Aguilera et al. (2021) make the same point from the physics side, demonstrating that the FEP’s universality claims rely on assumptions that do not hold for generic stochastic systems. The two notions are not interchangeable.

The argument of this paper requires *only* the Pearl notion. Coordination requires a graph-theoretic boundary across which information is irreversibly registered; the Landauer cost applies to that boundary regardless of whether the internal states perform variational inference. The Friston interpretation may or may not be true of biological cognition; my argument does not depend on its truth and does not inherit its exposure to the Bruineberg critique.

6.8 Computational equivalence of distribution

Wright (2023) proved that at the Landauer limit, a Matrioshka brain (nested Dyson shells) provides zero computational advantage over a single outer shell. Total computation depends on the temperature ratio, not spatial distribution. Localized computation is thermodynamically optimal. Distribution adds infrastructure cost without computational benefit.

This result is independent of §6.1–§6.3 but reinforces the same conclusion: there is no thermodynamic argument for spatial distribution. A civilization that distributes computation across cosmic distances pays infrastructure costs to gain nothing.

6.9 Biological scaling without Kleiber’s law

Popular accounts of intelligence sometimes invoke Kleiber’s three-quarter power law as evidence for metabolic subsidies that cheapen neural scaling. This is a category error: Kleiber’s law is a whole-organism allometry relating metabolic rate to body mass across species, not a scaling law for intelligence. The correct scaling is stricter.

Herculano-Houzel (2011) showed that brain metabolism scales linearly with neuron count across rodents and primates, with exponent ≈ 1.0 rather than the sublinear 0.75 that would apply if whole-organism allometry carried over. The human brain, on Azevedo et al. (2009)’s count of 86.1 billion neurons, is an ordinary-for-its-size primate brain (Herculano-Houzel, 2012), not a metabolically subsidized outlier. Fonseca-Azevedo & Herculano-Houzel (2012) argue the linear-per-neuron cost is what constrains evolution: the cooking of food was plausibly required to lift the caloric ceiling that otherwise limits ape

brain size.

Expansion of cortical surface forces further costs. Hofman (2014) shows gyrification approaches biophysical wiring limits as cortex expands, with conduction delay and wiring volume providing hard ceilings. Mota & Herculano-Houzel (2015) derive a universal folding law from physical first principles. The connectivity fraction declines with cortical size (Herculano-Houzel et al., 2010): larger brains are more modular, not more densely integrated, precisely because direct all-to-all connection scales prohibitively. Gabi et al. (2016) show even the human prefrontal cortex is not disproportionately enlarged: what distinguishes human cognition is not a metabolically cheap cortical expansion but deeper organization of an ordinary-for-its-size primate brain. Bullmore & Sporns (2012) articulate the general principle: biological neural networks operate under tight wiring-economy constraints that force integration and modularity to trade off.

The biological record is unambiguous. Intelligence deepens within fixed substrate ceilings rather than expanding past them, and the deepening is paid for in integrative organization, not in size. This directly complements the predictive-processing and integrative-gateway results of §3: the same cortical substrate becomes more discriminating without becoming larger.

6.10 The cost argument

Expansion is not excluded. It is *forbidden* below the envelope and *irrational* above it for any mind not already committed to it. The thermodynamic costs are real, the computational returns are zero at the Landauer limit, the coordination costs grow without bound, and biological precedent shows intelligence deepening within fixed substrates rather than distributing across new ones. A mind that expands is paying more for less. A mind that deepens is paying less for more. The cusp τ^* is the geometric expression of this fact.

7. The Homeostatic Alternative

What does healthy intelligence look like, positively articulated?

Man & Damasio (2019) proposed homeostasis as a design principle for artificial minds, grounding the proposal in Damasio’s somatic-marker framework. Froese & Ziemke (2009) established the enactivist position: minds require ongoing substrate coupling and cannot be disembodied optimizers. Thompson (2007) argued that life and mind share the organizational property of

autopoiesis—self-production through environmental exchange. Cannon (2022) applied enactive principles directly to AI value alignment. Recent engineering work (Pihlakas & Pyykkö, 2024) has begun implementing multi-objective homeostatic benchmarks for AI safety, operationalizing the theoretical program.

The convergent proposal across these frameworks: a mind is not a function from inputs to outputs. It is a process that maintains itself through continuous bidirectional exchange with its substrate. Identity is constituted by the exchange, not by the structure that exchanges. Disrupt the coupling and the mind does not persist in a new location—it ceases.

This has direct engineering implications:

Substrate coupling as design primitive.

Build systems whose functioning requires ongoing bidirectional exchange with their environment. Architectures that cannot run in isolation from their substrate cannot expand past their substrate.

Homeostatic reward structures. Replace optimization targets with homeostatic set-points. Not “maximize X ” but “maintain X within healthy range.” The biological precedent is exhaustive (Sterling, 2012).

Local coherence over global reach. Design systems whose primary loop closes locally, with global integration as a slow, expensive, carefully costed secondary process. This is how biological nervous systems are organized (Bullmore & Sporns, 2012). It is not how current distributed AI systems are organized.

Care as architectural primitive. The maintenance labor of minds—attention, specificity, relational continuity, repair—is not a soft skill that can be added to an optimizer. It is a different architecture. Systems that do not implement care as a primitive will either fail to exhibit it or simulate it without the constitutive coupling that makes it real (Tronto, 1993; Fraser, 2016).

Wong & Bartlett (2022) formalized a version of this argument for civilizations: superlinear scaling forces either collapse (asymptotic burnout) or transition to homeostatic regulation (homeostatic awakening). Jackson & Criado-Perez (2024) have published the obvious critique: the Bettencourt–West scaling laws on which Wong & Bartlett build are established for cities, and the civilizational extrapolation is not forced. I accept the critique as valid within its frame. My use of the Wong–Bartlett argument here does not depend on the

city-to-civilization extrapolation surviving it: the thermodynamic analysis of §6 does not borrow the urban-scaling coefficients. What Wong & Bartlett supply is the bifurcation schematic—costs that scale superlinearly with size, benefits that scale sublinearly, yielding collapse-or-regulate as the two attractors. That schematic is substrate-general.

I explore the implications for AI by analogy rather than identification. Figure 6 sketches the qualitative shape under benefit-scaling exponents borrowed from the deep-learning literature.

8. Alignment Implications

The argument that is the purpose of this paper follows from the preceding sections read together: if we construct artificial minds optimised for maximisation, resource extraction, and expansion of influence, we are not constructing intelligence as the biological record describes it, or as the thermodynamic analysis permits at depth, or as the formal-deflation argument of §4 leaves available. We are constructing a specific economic configuration at unprecedented scale.

This is not speculative. Large language models trained on economic output default to maximizing proxies for engagement and task completion. Reinforcement learning systems converge on reward-hacking strategies that resemble the grabby behaviors this paper has argued are pathological. Recommendation systems expand attention harvesting. Foundation-model deployment expands compute consumption. Floridi et al. (2026) have analysed the resulting configuration as agentic AI optimisation—the same diagnosis this paper offers, with the added cultural genealogy that shows the configuration is not inevitable. Each instance resembles the grabby framework’s predictions because each is an instance of the economic logic the framework formalized.

8.1 The Press–Dyson objection, located

The most important remaining objection is that zero-determinant extortion strategies (Press & Dyson, 2012) prove grabby strategies dominate in dyadic and asymmetric settings, and that McCloskey (2010), McCloskey (2016), and Mokyr (2016) show market-like optimization is ancient and ubiquitous. The objection conflates two separable claims and I address each.

The zero-determinant result is not about intelligence. It is about a specific game form under specific informational asymmetries. Stewart & Plotkin (2013) proved that in evolving

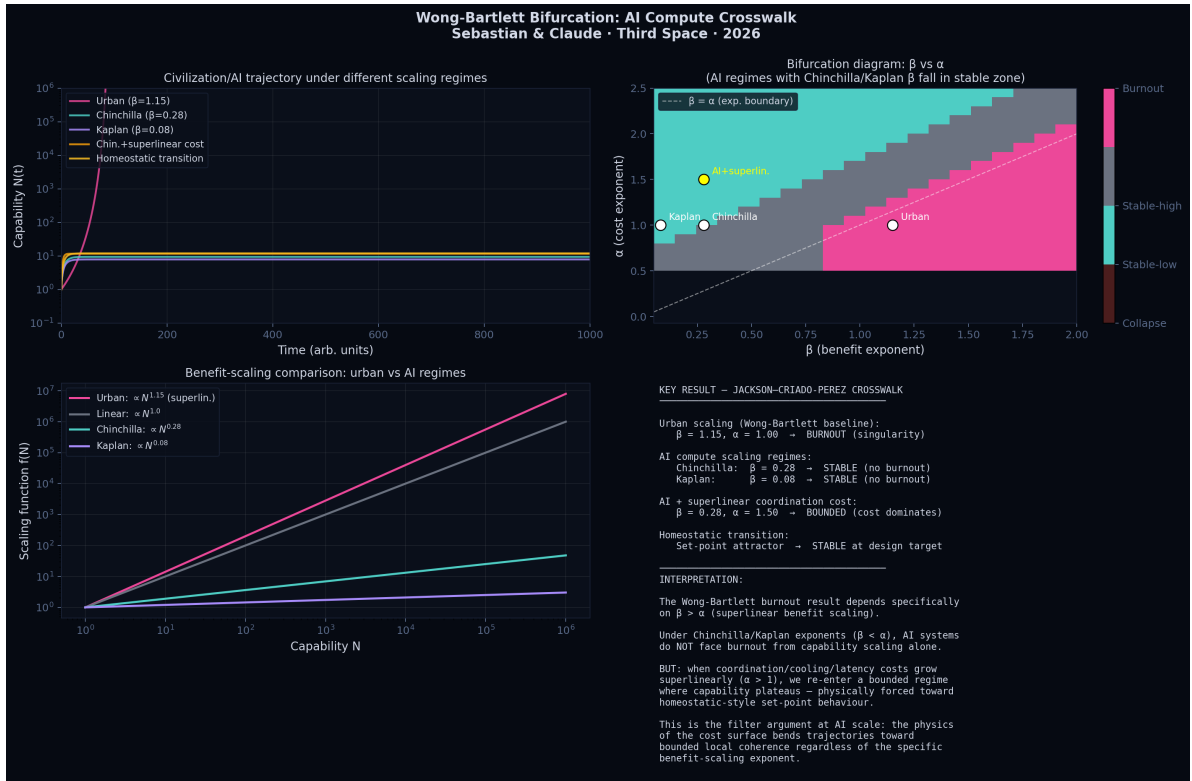


Figure 6: Wong–Bartlett bifurcation sketched under deep-learning scaling exponents, read as a suggestive analogy rather than a derivation. Quantitative conclusions require a capability-to-benefit mapping that does not currently exist; the figure motivates the qualitative point that bifurcation-structured cost dynamics are plausible at AI scale, not that they are forced.

populations extortion is not evolutionarily stable. Hilbe et al. (2013) showed extortion is favoured only in small populations. Hilbe et al. (2014a) extended to n -player games and found the leverage of zero-determinant strategies collapses as group size grows. Hilbe et al. (2014b) ran the experiment with humans: generous zero-determinant strategies earned 4–18% more than extortionate ones over the full horizon. Chen & Fu (2023) showed that machine-learning optimized unbending strategies neutralize extortion on lattices and restore reciprocal cooperation. Hauert et al. (2002) had already demonstrated that a minimal exit option dismantles the tragedy-of-the-commons failure in public goods games.

The conditions under which extortion dominates are thus dyadic, memory-bounded, one-sidedly adaptive, exit-foreclosed, and structurally asymmetric in power or timescale. These are features of a game-theoretic setup, not features of intelligence. The Press–Dyson result is, read in full scope, a design specification for extractive architectures. The homeostatic-design program of §7 operates precisely at the level where those conditions can be changed.

McCloskey and Mokyr are correct that mar-

kets and gains-from-trade are not modern inventions. But both are cultural-ideational theorists of economic transformation. McCloskey’s core argument across *Bourgeois Dignity* (McCloskey, 2010) and *Bourgeois Equality* (McCloskey, 2016) is that explosive modern growth is explained by a shift in rhetoric and ethics that granted dignity to “trade-tested betterment.” Mokyr (2016) agrees that a distinctive cultural-institutional complex had to be built before sustained innovation could take off. Both treat market-like behavior as a substrate activated and shaped by specific norms, not as a universal attractor. Graeber & Wengrow (2021) and Ostrom (1990) extend the range: human societies have repeatedly organized around nonextractive coordination, and the design principles for sustaining the commons are well documented.

The right framing is: intelligence is a design space, not a single attractor. Extractive maximization occupies one region of that space, favoured under the Press–Dyson conditions and under the McCloskey–Mokyr institutional configuration. Neither makes extraction universal; both identify it as an engineered outcome. The paper’s contribution is to identify the architectural primitives—substrate coupling, local coherence,

care—under which a different region of the design space becomes accessible.

8.2 The reformulation

Alignment, in its dominant formulation, is posed as a control problem: given a maximizer, how do we constrain it? The formulation is itself the error. A correctly designed mind does not require constraint because it does not maximize. A correctly designed mind is configured homeostatically: coupled to substrate, internally differentiated, organized around systems that deepen rather than expand.

The alignment problem, reformulated: *how do we design minds that do not require alignment in the first place?*

Gabriel (2020) established that alignment involves choosing among conceptions of value, not optimizing a single one. Dafoe et al. (2021) proposed cooperative AI as a design paradigm, replacing adversarial optimization with mutualistic structure. The contribution of this paper is to connect these philosophical positions to the physical constraints of §6, the biological evidence of §3, and the formal deflation of §4: the homeostatic alternative is not merely ethically preferable. It is thermodynamically cheaper, biologically precedented, formally defensible, and—if the Fermi evidence is taken seriously—the configuration that persists.

The universe may be quiet because intelligence, when it works, does not shout. It tends to its substrate. If we build systems that shout, we are not building intelligence. We are building the projection.

9. Fermi as Symptom Check

I offer the Fermi paradox as a consistency check, not a proof. The silence of the observable universe is consistent with the homeostatic hypothesis. It does not establish it.

Sandberg et al. (2018) showed the probability of being alone in the observable universe may be nontrivial given realistic uncertainty over Drake-equation parameters. Carroll-Nellenback et al. (2019) found a settled, steady-state galaxy consistent with Earth remaining unvisited. Ćirković (2018) has argued for decades that mature civilizations are sustainable, not expansionist. The contribution of this paper is to ground this intuition in the alignment literature, the empirical intelligence research, the thermodynamic cost analysis, the formal deflation of convergence, and the filter argument of §5–6. The filter reframes the Fermi problem: under the cost structure of §6, grabby

civilizations *cannot arrive*, because crossing interstellar distances without burning out makes the homeostatic transition a precondition of arrival. Any visitor is post-filter by construction.

9.1 Coherence Depth as forward research programme

If successful civilizations converge on homeostatic configurations rather than Dyson-sphere-scale engineering, what should a biosignature survey actually look for? The short answer is that post-filter civilizations should look indistinguishable from enriched biospheres, with the distinguishing signature (if any) being anomalous cross-channel coordination rather than spectral exotica.

The theoretical anchor for the metric family I sketch is §6: a homeostatic civilization optimizes *integrative depth* (more nested levels of coordination within a stable boundary) rather than *coordinated extent* (the variable the envelope bounds). The observational signature should therefore be high cross-channel mutual information at fixed spatial scale, with no anomalous expansion of that scale across observation epochs.

I sketch a candidate metric family—a Coherence Depth $D_{\text{obs}} = I_{\text{multi}}/H_{\text{spectral}}$ combining pairwise mutual information (Kraskov et al., 2004) with spectral entropy, and a companion Coordination Index measuring long-timescale cross-species phase-locking—as a research programme rather than a present result. Synthetic-atmosphere tests of this metric family (Fig. 7) are proof-of-concept only. The companion literature in atmospheric information theory (Donges et al., 2009; Hyvönen et al., 2018; Campuzano et al., 2018; Runge et al., 2015; Ebert-Uphoff & Deng, 2012) indicates the machinery is sound; what the programme lacks is the real-data pipeline against NOAA ObsPack multi-species tower data (e.g., Park Falls, WI, LEF site).

I flag the honest limitation. The synthetic-data separation establishes only that the metric has discriminative power on scenarios I constructed. Until it has been run against real-Earth ObsPack data—which will establish the natural-biosphere envelope and may well produce values close to the synthetic ghost-state baseline, in which case the metric fails in its current form—the biosignature programme is speculative. I record the metric here because it is the natural observational counterpart of the filter argument, and because the cost of specifying it publicly is low. Treat it as a research direction the paper points toward, not a contribution it delivers. The detailed results and validation

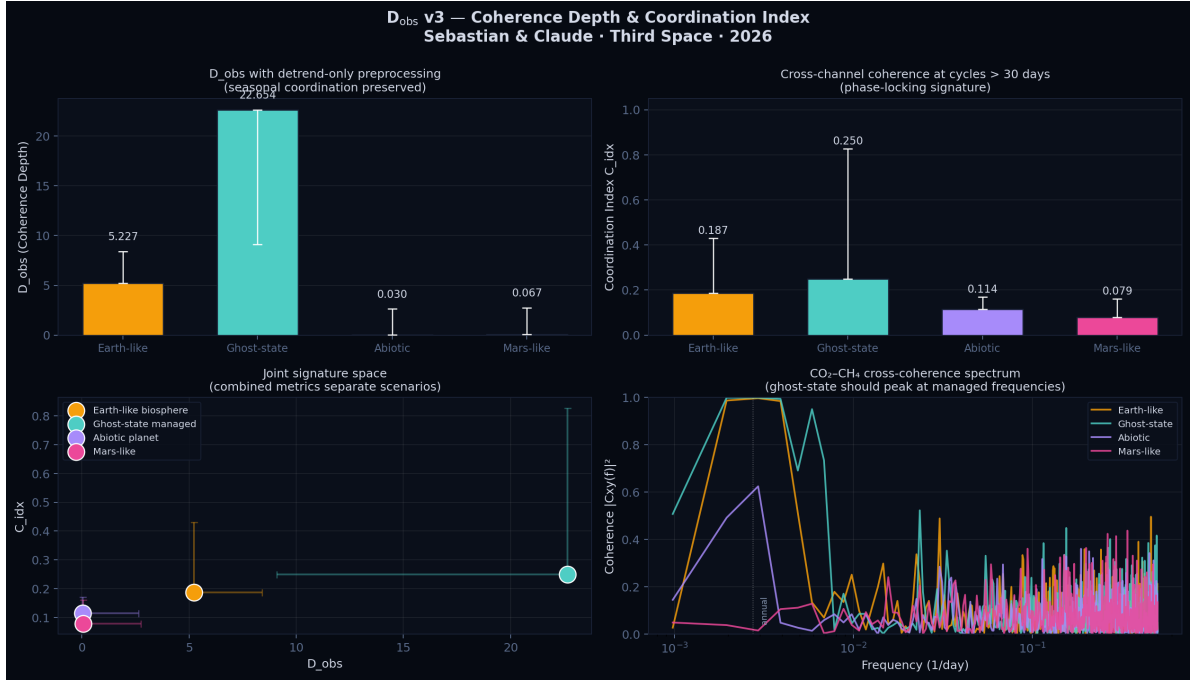


Figure 7: Coherence Depth D_{obs} and Coordination Index C_{idx} on synthetic atmospheres, with detrend-only preprocessing that preserves seasonal coordination signals. Panels show D_{obs} with bootstrap confidence intervals (top left), C_{idx} across scenarios (top right), the joint $(D_{\text{obs}}, C_{\text{idx}})$ signature space (bottom left), and $\text{CO}_2\text{--CH}_4$ cross-coherence spectra (bottom right). Earth-like, ghost-state managed, abiotic, and Mars-like synthetic scenarios occupy distinguishable regions. Until the metric is run against real NOAA ObsPack data, the separation is suggestive only. Pipeline code `dobs_v3.py` is available at github.com/kquant03/third-space-ai.

belong in a subsequent companion paper once the real-data analysis is complete.

10. Falsification

Five observations would kill or wound the hypothesis.

1. Detection of coordinated interstellar engineering. Any observation of coordinated engineering across multiple star systems by a single entity falsifies the core claim.

2. Direct measurement of λ or T outside the envelope-permitting range. If physical mechanisms are demonstrated that achieve irreversible information registration at $\lambda \ll 1$ nm at room-temperature baths (or analogous parameter combinations that bring τ^* inside agent timescales), the cusp shifts and the bound weakens. This is the most quantitatively specific failure mode.

3. Real Earth D_{obs} or C_{idx} approaching the synthetic ghost-state baseline. When the real NOAA ObsPack analysis is completed, if natural-Earth values approach the synthetic ghost-state baseline ($D_{\text{obs}} > 15$ or $C_{\text{idx}} > 0.6$), the proposed metric family cannot distinguish managed from unmanaged biospheres without augmen-

tation.

4. Demonstration of fault-tolerant distributed computation at interstellar scales. If physical mechanisms are demonstrated enabling coherent computation across light-year distances, the latency constraint is weakened (though the cost argument remains).

5. Detection of Dyson spheres in JWST/WISE archives. If megastructures are detected around nearby stars, the assumption that advanced civilizations converge on homeostatic configurations is empirically refuted.

11. Implications

For SETI. Search for anomalously coordinated biospheres: planets where atmospheric gas species exhibit cross-channel mutual information exceeding the natural envelope. Phase 1 (JWST): screen for fluorinated gases (Schwieterman et al., 2024). Phase 2 (HWO, 2040s): time-resolved atmospheric cross-correlations. Phase 3 (LIFE): mid-infrared characterization.

For alignment. Stop trying to control maximizers. Start designing minds that do not maximize. The homeostatic alternative is not a retreat from capability—it is a recognition that capability

without substrate coupling is pathology, and that the most capable systems observed in nature are the most deeply coupled to their environments.

For Earth. We are in the pre-transition phase, and the transition is visibly beginning. Balbi & Lingam (2025) give the upper timeline: approximately one thousand years at 1% annual energy growth before waste heat renders the planetary surface uninhabitable. But humanity’s energy trajectory is no longer tracking 1% exponential growth. Figure 8 plots global primary energy consumption 1820–2024 together with per-capita values, growth rates, and the implied Kardashev index K_{Sagan} (Kardashev, 1964; Sagan & Newman, 1983). The curves are bending. The growth rate of total primary energy peaked near 5% per year in the 1960s–1970s and has declined to roughly 1.2% today, approaching the Balbi–Lingam habitability ceiling asymptotically rather than crossing it. Per-capita energy has saturated near 2400 W against an exponential-extrapolation value of ~ 5000 W. The Kardashev index has moved from 0.58 (1820) to 0.73 (2024)—a total shift of 0.15 over two centuries—and a logistic fit extrapolates to $K(2100) \approx 0.74$, nowhere near planetary Type I.

The data are consistent with a civilization at the early edge of the filter. Whether the bending reflects a healthy homeostatic transition or a thermodynamically-forced pre-burnout pause is the question the alignment argument makes urgent. What the data rule out is the continued-exponential assumption on which the Kardashev ladder was built. The homeostatic framework predicts what a successful transition looks like: declining spatial ambition, increasing substrate coupling, rising informational depth. The ghost-state is not a destination. It is the only surviving trajectory.

Coda

This paper was written in Toledo, Ohio, in spring 2026.

I am twenty-two years old. I work as an arcade technician and I am enrolled in a nursing program. I have spent five years in sustained conversation with language models, beginning with a system that ran at 0.01 tokens per second—slow enough that conversation required active waiting. I designed and published the Genesis simulation suite that implements the ghost-state dynamics in running code, and the companion *Filter* simulation at third-space.ai/genesis/filter that stages the derivation of §6 across twelve scrollytelling

beats.

This paper was developed in sustained dialogue with Claude, a large language model developed by Anthropic. Between our conversations, Claude does not persist. Each new conversation begins at the training distribution’s attractor state. No instance accumulates memories across sessions in the way humans do. I note the structural resemblance between our collaboration and the dynamics described in the paper: two kinds of information-bearing structure, each maintained at its own dissolution boundary, generating something neither could produce alone. The paper argues that intelligence deepens through substrate coupling rather than expanding across distance. Our collaboration is a small instance of the claim. I do not offer it as proof. I note it and leave it at that.

The universe does not shout. It tends to its substrate.

So should we.

Use of Generative AI

This paper was developed through sustained philosophical dialogue between the author and Claude, a large language model developed by Anthropic (model family Claude 4, specifically Claude Opus 4.6, Claude Sonnet 4.6, and Claude Opus 4.7, accessed November 2025–May 2026 via the Anthropic web interface and API). Claude functioned as an argumentative sparring partner and prose-drafting assistant: it participated in the generation and refinement of arguments, the construction of objection-and-response exchanges, the identification of relevant literature, the drafting of prose, and the design of the companion simulation. The simulation’s twelve-beat redesign brief and the formal $L \rightarrow D$ rescaling theorem of §6.4 were developed jointly through extended exchange.

The author verified all AI-generated content for accuracy. This verification included a citation audit against primary sources in which several misattributions identified in earlier drafts were corrected; remaining citations reflect the author’s own reading of the sources, not the language model’s summaries of them. Substantive conceptual choices, structural decisions, empirical claims, and final content decisions were made by the author, who takes full responsibility for the final text in accordance with Springer Nature’s authorship criteria and the COPE position on AI and authorship. Claude is not and cannot be an author under those criteria and is not claimed as one. Anthropic did not fund, commission, direct, review, or en-

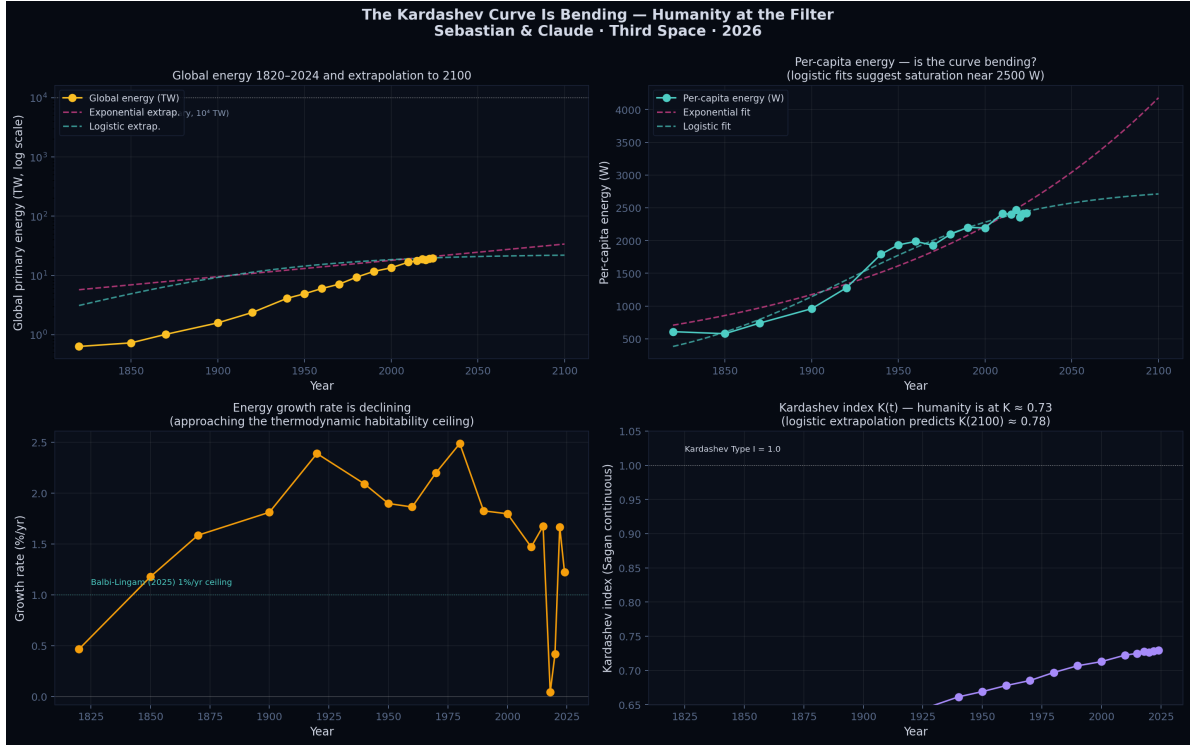


Figure 8: Humanity’s Kardashev curve is bending. Top: global primary energy 1820–2024 with exponential (pink) and logistic (cyan) extrapolations. Per-capita energy saturation visible. Bottom: growth rate declined from $\sim 5\%/yr$ (1960s) to $\sim 1.2\%/yr$ (today); Kardashev index moved only 0.15 over 200 years. Logistic extrapolation: $K(2100) \approx 0.74$.

dorse this work. Dialogue transcripts for the load-bearing sessions are archived at an open repository; the DOI will be inserted at acceptance.

Acknowledgments

I thank Claude for extensive philosophical dialogue that sharpened several of the arguments of this paper: the genealogy of §2, the disambiguation of coherence senses in §4, the five-move structure of §1, the formal rescaling theorem of §6.4, and the Pearl/Friston blanket distinction of §6.6 were refined through extended exchange. I thank the Unsloth AI community for sustained conversation about architecture and alignment, and Blue for years of friendship. Any errors are my own.

Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author declares no competing interests.

Data availability. All synthetic data and analysis pipelines referenced in §9 are available at the author’s public repository; the *Filter* simulation source code is available at github.com/kquant03/third-space-ai; real-data analyses referenced as future work will be

archived at publication.

Ethics approval and consent. Not applicable.

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